

Ankit Suresh Raut | Software Engineer

Fairfax, VA (Open to Relocation) | +1 571-224-4890 | ankitsureshraut@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Software Engineer with **3.5+ years of experience** designing, developing, and optimizing backend services, distributed systems, and cloud-native applications. Strong expertise in **Java, Spring Boot, Go, Python, REST APIs, microservices, AWS, Kafka, SQL, Docker, Kubernetes, and CI/CD pipelines**. Experienced in building scalable APIs, event-driven workflows, fault-tolerant services, and high-availability systems supporting large-scale transaction and workflow processing. Proven ability to improve system reliability, reduce latency, optimize cloud infrastructure, strengthen observability, and deliver production-ready software in Agile engineering environments.

Skills

Programming Languages: Java, Go, Python, SQL
Backend Development: Spring Boot, REST APIs, Microservices, Distributed Systems, API Versioning, Service-Oriented Architecture
Cloud & DevOps: AWS Lambda, API Gateway, RDS, DynamoDB, S3, CloudWatch, Docker, Kubernetes, Terraform, AWS CDK, CI/CD
Messaging & Streaming: Kafka, AWS SNS/SQS, Event-Driven Architecture, Asynchronous Processing
Databases: PostgreSQL, MongoDB, DynamoDB, Oracle
Testing & Quality: JUnit, pytest, Integration Testing, Testcontainers, Unit Testing, Code Reviews
Monitoring & Reliability: Structured Logging, Observability, Distributed Tracing, Metrics, Fault Tolerance, High Availability
Architecture Concepts: Idempotency, Retry Logic, Circuit Breaker Pattern, Concurrency, Scalability, Design Patterns
Methodologies: Agile, Scrum, SDLC, Sprint Planning, Cross-Functional Collaboration

Work Experience

PTC, Boston, MA (Remote) | Software Engineer

Feb 2026 - Present

- Designed and developed **Go-based cloud-native microservices** supporting distributed workflow execution, service coordination, and high-availability backend operations.
- Built scalable backend APIs using **AWS Lambda, API Gateway, RDS, DynamoDB, S3, and CloudWatch**, improving platform reliability and supporting **99%+ service uptime**.
- Implemented asynchronous processing patterns, retry mechanisms, and fault-handling workflows to improve system resilience across distributed services.
- Standardized REST API contracts, versioning strategies, and backward compatibility practices, reducing integration issues during production releases.
- Enhanced CI/CD deployment workflows using automated validation and blue-green deployment practices, reducing production incidents by **40%**.
- Improved production observability through structured logging, metrics, CloudWatch monitoring, and distributed tracing, reducing issue diagnosis and MTTR by **30%**.
- Optimized AWS infrastructure usage, service configuration, and resource allocation, reducing monthly cloud spend by approximately **\$800-\$1,000**.

Citi, New York, NY | Software Engineer

Jan 2025 - Sept 2025

- Developed and maintained **Java Spring Boot microservices** for financial transaction processing, payment validation, and secure backend workflow automation.
- Designed RESTful APIs with request validation, structured exception handling, authentication controls, and audit-focused logging to support enterprise security standards.
- Integrated Kafka-based event streaming to enable real-time transaction updates, asynchronous service communication, and downstream system processing.
- Implemented idempotency keys, retry logic, and failure recovery patterns, improving transaction reliability and reducing duplicate processing risks.
- Optimized SQL queries, indexing strategies, and database access patterns, reducing transaction response latency by **25%** under high-volume workloads.
- Strengthened application resilience using circuit breaker patterns, centralized logging, service monitoring, and production support best practices.

LTIMindtree, Pune, India | Software Engineer P1

Sept 2022 - Dec 2023

- Engineered **Spring Boot microservices** supporting enterprise applications and backend platforms handling **2M+ API requests per month** with **98% uptime**.
- Developed REST APIs for transaction processing, data exchange, and service integration, supporting **30K+ daily transactions** across business workflows.
- Implemented API versioning, rate limiting, request validation, and structured exception handling to improve API reliability and maintainability.
- Improved distributed event throughput by **30-35%** using **Kafka, AWS SNS/SQS, and asynchronous processing patterns**.
- Reduced production defects by **40%** by strengthening integration testing, improving failure-handling logic, and participating in peer code reviews.
- Tuned SQL queries, indexes, and database interaction patterns to improve response times under concurrent user load.
- Worked in Agile development cycles with cross-functional teams to deliver backend enhancements, resolve defects, and improve system scalability.

Zentek Infosoft, India | Software Developer

May 2021 - Aug 2022

- Developed backend modules using **Java, Spring Boot, REST APIs, and PostgreSQL** for enterprise workflow automation applications.
- Built reusable API components for user workflows, data processing, reporting, and internal system integrations.
- Assisted in Docker containerization and cloud migration initiatives to improve deployment consistency across development and production environments.
- Improved backend performance by optimizing database queries, refining service logic, and reducing API response latency.
- Wrote unit tests, supported integration testing, and resolved application defects to improve code quality and production stability.
- Participated in code reviews, sprint planning, and design discussions to enforce clean code, design patterns, and reliability standards.

Education

Master of Science, Computer Science	Jan 2024 - Dec 2025
George Mason University, Fairfax, Virginia	GPA: 3.67/4.0
Bachelor of Engineering, Computer Engineering	Jun 2018 - Apr 2022
Pune University, India	GPA: 9.30/10

Projects

Backend Orchestration & Workflow Platform

- Designed a distributed backend orchestration service to coordinate multi-stage workflows across microservices using asynchronous execution and event-driven processing.
- Implemented task scheduling, workflow status tracking, retry-safe execution, and failure recovery logic, achieving sub-2s latency under concurrent workloads.
- Improved service reliability by applying idempotency, structured logging, database optimization, and containerized deployment practices.

Voice-to-Insights Distributed Pipeline

- Built an event-driven backend pipeline to process streaming audio data into searchable real-time insights using scalable REST APIs and asynchronous ingestion.
- Developed processing workflows for data ingestion, transformation, indexing, and retrieval, enabling horizontal scaling under continuous workloads.
- Improved pipeline performance through queue-based processing, structured logging, and optimized API response handling.

AWS Multi-Account Landing Zone (CDK + Control Tower)

- Engineered an infrastructure-as-code platform using **AWS CDK and Control Tower** to automate provisioning of secure multi-account AWS environments.
- Implemented governance guardrails, IAM controls, centralized logging, and observability standards to improve cloud security and operational scalability.
- Reduced manual infrastructure setup by automating account creation, baseline configuration, and compliance-focused cloud resource provisioning.

Certifications

- AWS Certified Developer – Associate (In Progress)
- Oracle Cloud Infrastructure AI Foundations Associate - 2023
- Oracle Cloud Infrastructure Data Science Professional - 2023
- Oracle Cloud Digital Assistant Professional - 2023
- Java Programming Masterclass (Java 11 & 17)